

03/11/2026

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Spectral algebraic

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geometry

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Last time More on the functor of points, quotients

Today Begin Spectral algebraic geometry

Summary from last time

(1) The functor of points of a scheme satisfies fpqc descent

(2) We can define  $\mathrm{QCoh}(-)^{\heartsuit} : \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\heartsuit} \rightarrow \mathrm{CMon}(\mathrm{Cat}_1)$   
by extending  $\mathrm{Mod}_{(-)}^{\heartsuit} : \mathrm{CAlg} \rightarrow \mathrm{CMon}(\mathrm{Cat}_1)$

$$\mathrm{QCoh}(X)^{\heartsuit} := \lim_{\mathrm{Spec}(A) \rightarrow X} \mathrm{Mod}_A^{\heartsuit}$$

(3)  $\mathrm{QCoh}(-)^{\heartsuit}$  satisfies fpqc descent

Origin of the theory of stacks If we're given a scheme  $X$  with an action of a group scheme  $G$ , the quotient  $X/G$  in  $\mathrm{Fun}_{\mathrm{fpqc}}(\mathrm{CAlg}, \mathrm{Set})$  may not be very interesting / not capture equivariant geometry.

- > If  $X = *$ , then the quotient is always  $*$
- > The image of  $\mathbb{A}^{n+1}/G_m$  at the origin should be interesting, but isn't
- > Basically we want an analogue of the topologist's  $BG$  that classifies  $G$ -bundles
  - The problem is that while the groupoid of  $G$ -bundles forms a sheaf, the set of isomorphism classes of  $G$ -bundles does not
  - So we should enlarge our world to  $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{Gpd})$   
 Note that  $\text{Gpd} \simeq \text{An}_{\leq 1}$
- > But the problem persists. We should enlarge to  $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{An}_{\leq 2})$ , then  $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{An}_{\leq 3})$ , ..., then  $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{An})$

## Spectral algebraic geometry beginnings

So far. We have the symmetric monoidal stable  $\infty$ -category  $(\mathcal{S}_p, \otimes, \mathbb{S})$ .

> It is also reasonable to call this  $\text{Mod}_{\mathbb{S}}$

> We have the free  $\mathbb{S}$ -module functor

$$\mathbb{S}\langle - \rangle := \sum_{+}^{\infty} : (A_n, \wedge) \rightarrow (\text{Mod}_{\mathbb{S}}, \otimes)$$

which is symmetric monoidal.

> We can talk about commutative algebras in  $\text{Mod}_{\mathbb{S}}$ , as well as commutative coalgebras.

- This is actually more technical than the case of commutative monoids with respect to cartesian product.

- We won't go into the details, but you can assume it works as expected.

- > So to formulate non-affine toric geometry, we need an analogue of scheme theory
- Before doing so, let us explain the relationship between classical ring theory and commutative algebras in  $\mathcal{S}p$ .

## Comparison to Classical algebra

Thm. The unique colimit-preserving functor  $Sp \rightarrow D(\mathbb{Z})$  determined by  $S \mapsto \mathbb{Z}$  refines to a symmetric monoidal functor

$$(Sp, \otimes) \longrightarrow (D(\mathbb{Z}), \otimes_{\mathbb{Z}}^{\mathbb{L}})$$

We denote its right adjoint by  $H: D(\mathbb{Z}) \rightarrow Sp$  and call this the Eilenberg-MacLane functor.

Remark. The name is because  $H\mathbb{Z}$  is the spectrum representing homology with coefficients in  $\mathbb{Z}$ .

Intuitionistic definition. If  $\mathcal{C}$  and  $\mathcal{D}$  are sym mon  $\infty$ -categories, there is a weakening of the notion of a symmetric mon. functor  $F: \mathcal{C} \rightarrow \mathcal{D}$  called a lax symmetric monoidal functor, which specifies a map

$$1_{\mathcal{D}} \longrightarrow P(1_{\mathcal{C}})$$

and maps

$$F(X_1) \otimes_D \cdots \otimes_D F(X_n) \longrightarrow F(X_1 \otimes_e \cdots \otimes_e X_n)$$

subject to all higher coherences

Fact. If  $F: \mathcal{C} \longrightarrow \mathcal{D}$  is a lax symmetric monoidal functor, then  $F$  induces a functor  $\text{CAlg}(\mathcal{C}) \longrightarrow \text{CAlg}(\mathcal{D})$  fitting into a commutative square

$$\begin{array}{ccc} \text{CAlg}(\mathcal{C}) & \longrightarrow & \text{CAlg}(\mathcal{D}) \\ \text{forget} \downarrow & & \downarrow \text{forget} \\ \mathcal{C} & \xrightarrow{F} & \mathcal{D} \end{array}$$

Fact. If  $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$  is an adjunction, then a Sym. monoidal structure on the left adjoint  $F$  induces a lax Symmetric monoidal structure on the right adjoint  $G$ .

Upshot The Eilenberg-MacLane functor  $H: D(\mathbb{Z}) \rightarrow Sp$  is lax sym mon and induces a functor

$$H: \text{CAlg}(D(\mathbb{Z})) \rightarrow \text{CAlg}(Sp)$$

Observe The inclusion  $\text{Mod}_{\mathbb{Z}}^{\heartsuit} \hookrightarrow D(\mathbb{Z})$  as the complexes concentrated in degree 0 is lax symmetric monoidal. This is because

$$H_0(M_1[0] \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n[0]) \cong M_1 \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n$$

> There's a natural map from a connective chain complex to its  $H_0$ , which gives the desired map

$$M_1[0] \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n[0] \longrightarrow M_1 \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n$$

> Hence we obtain functors

$$\left\{ \begin{array}{c} \text{comm.} \\ \text{ring} \end{array} \right\} = \text{CAlg}(\text{Mod}_{\mathbb{Z}}^{\heartsuit}) \hookrightarrow \text{CAlg}(D(\mathbb{Z})) \xrightarrow{H} \text{CAlg}(Sp)$$

⚠ If  $A$  and  $B$  are ordinary commutative rings, then there is a map

$$H_A \otimes_S H_B \longrightarrow H(A \otimes_{\mathbb{Z}} B)$$

that induces an isomorphism on  $\pi_0$ , but generally  $H_A \otimes_S H_B$  has higher homotopy.

Note. If  $\mathcal{C}$  is a sym mon  $\infty$ -category and  $A \in \text{CAlg}(\mathcal{C})$ , one can talk about the  $\infty$ -category  $\text{Mod}_A(\mathcal{C})$  of (left)  $A$ -modules. This comes equipped with a forgetful functor  $\text{Mod}_A(\mathcal{C}) \rightarrow \mathcal{C}$  and a left adjoint

$$A \otimes_{\mathcal{C}} (-) : \mathcal{C} \longrightarrow \text{Mod}_A(\mathcal{C}).$$

Thm. If  $A$  is an ordinary commutative ring, then the unique colimit-preserving functor  $\text{Sp} \rightarrow D(A)$  given by  $S_1 \mapsto A$  induces an equivalence

$$\text{Mod}_{H_A}(\text{Sp}) \xrightarrow{\sim} D(A).$$

> Moreover, this equivalence refines to an equivalence of symmetric monoidal  $\infty$ -categories.

Convention For an ordinary ring  $A$ , we simply write  $A$  for  $HA \in \text{CAlg}(\text{Sp})$ , and  $\text{Mod}_A$  for

$$\text{Mod}_{HA}(\text{Sp}) \cong D(A).$$

## The graded ring structure on homotopy groups

Def. A commutative algebra in spectra is called an  $\mathbb{E}_\infty$ -ring spectrum or  $\mathbb{E}_\infty$ -ring. An  $\mathbb{E}_\infty$ -ring  $A$  is connective if the underlying spectrum of  $A$  is connective, i.e.,  $\pi_n(A) = 0$  for  $n < 0$ .

Notation. From now on, we write  $\mathbf{CAlg} = \mathbf{CAlg}(\mathcal{S})$  and we write  $\mathbf{CAlg}^\heartsuit = \mathbf{CAlg}(\mathbf{Mod}_{\mathbb{Z}}^\heartsuit)$  for the ordinary category of commutative rings. ↙ should be bold

> We write  $\mathbf{CAlg}^{\text{cn}} \subset \mathbf{CAlg}$  for the full subcategory spanned by the connective  $\mathbb{E}_\infty$ -rings.

Observe. Let  $E$  be a spectrum. Then since  $\mathcal{S} = \sum_{+}^{\infty} *$ , we see that

$$\begin{aligned} \mathrm{Map}_{\mathcal{S}}(\mathcal{S}, E) &\cong \mathrm{Map}_{\mathcal{A}n}(*, \Omega^\infty E) \\ &\cong \Omega^\infty E. \end{aligned}$$

> In particular,  $\pi_0 \text{Map}_{\text{Sp}}(S, E) \simeq \pi_0(E)$  as we defined before

> Similarly,

$$\begin{aligned} \text{Map}_{\text{Sp}}(S[n], E) &\simeq \text{Map}(S, E[-n]) \\ &\simeq \Omega^\infty E[-n] \end{aligned}$$

This is the anima we earlier denoted  $E_{-n}$ . So we see that  $\pi_n(E)$  as defined before is  $\pi_0 \text{Map}_{\text{Sp}}(S[n], E)$

Fact. We have  $\pi_0(S) \simeq \mathbb{Z}$ . This follows from the unstable computation  $\pi_n(S^n) \simeq \mathbb{Z}$ .

Topological exercise. For  $m, n \geq 0$ , the map of topological spaces

$$S^{m+n} \simeq S^m \wedge S^n \xrightarrow{\text{swap}} S^n \wedge S^m \simeq S^{m+n}$$

is homotopic to  $(-1)^{mn} \text{id}_{S^{m+n}}$ .

> Since  $\Sigma^\infty : (\text{An}, \wedge) \rightarrow (\text{Sp}, \otimes)$  is symmetric monoidal, we deduce that for  $m, n \geq 0$ ,

$$S[m+n] \simeq S[m] \otimes S[n] \xrightarrow{\text{swap}} S[n] \otimes S[m] \simeq S[m+n]$$

is  $(-1)^{mn} \text{id}$ . By shifting, this follows for all  $m, n \in \mathbb{Z}$ .

Def. A graded ring is an associative ring  $A$  together with a decomposition of abelian groups

$$A = \bigoplus_{d \in \mathbb{Z}} A_d$$

Such that

$$(1) A_m A_n \subset A_{m+n}$$

$$(2) \text{ Koszul sign rule: If } a \in A_m \text{ and } b \in A_n, \text{ then } ba = (-1)^{mn} ab.$$

Observe If  $A$  is a graded ring, then  $A_0$  is given the structure of a commutative ring. Moreover each graded piece  $A_d$  is provided with the structure of an  $A_0$ -module.

Construction. Let  $A$  be an  $E_\infty$ -ring with unit  $u: S \rightarrow A$  and multiplication  $\text{mult}: A \otimes A \rightarrow A$ . Write

$$\pi_*(A) := \bigoplus_{d \in \mathbb{Z}} \pi_d(A).$$

Define a graded ring structure on  $\pi_*(A)$  as follows.

> The unit is given by

$$\mathbb{Z} \simeq \pi_0(S) \xrightarrow{\pi_0(u)} \pi_0(A) \subset \pi_*(A)$$

> The multiplication  $\pi_m(A) \times \pi_n(A) \rightarrow \pi_{m+n}(A)$  is given by applying  $\pi_0$  to

$$(f, g) \longmapsto f \otimes g$$

$$\text{Map}(S[m], A) \times \text{Map}(S[n], A) \longrightarrow \text{Map}(S[m] \otimes S[n], A \otimes A)$$

$$\downarrow \text{mult}_*$$

$$\text{Map}(S[m+n], A)$$

> The Koszul Sign rule Comes from the earlier mentioned fact about the swap map.

> In particular,  $\pi_0(A)$  is a commutative ring

Def. A map of  $E_\infty$ -rings  $f: A \rightarrow B$  is faithfully flat if:

(1)  $\pi_0(f): \pi_0(A) \rightarrow \pi_0(B)$  is a faithfully flat map of ordinary rings

(2) For each  $n \in \mathbb{Z}$ , the induced map of  $\pi_0(B)$ -modules

$$\pi_n(A) \otimes_{\pi_0(A)} \pi_0(B) \rightarrow \pi_n(B)$$

is an isomorphism